MILITARY SPECIFICATION

RUBBER STOCK, UNVULCANIZED: FOR TRACK SHOES, PADS, AND PIN BUSHINGS

1. SCOPE

1.1 **Scope.** This specification covers unvulcanized rubber stock used for reconditioning rubberized track shoe assemblies for use on military tracked vehicles (see 6.1).

1.2 **Classification.** The unvulcanized rubber stock shall be of the following styles, as specified (see 6.2):

Style I Block stock
Style II Pin bushing stock

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L–P–378 —Plastic Film (Polyethylene Thin Gage)

MILITARY

JAN–P–100 —Packaging and Packing for Overseas Shipment, General Specification

MIL–P–116 —Preservation, Methods of

MIL–P–14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of

MIL–B–43014 —Boxes: Water Resistand Paperboard; Folding, Set-Up, and Metal-Stayed

MIL–V–45349 —Vehicles: Combat and Special Purpose; Tactical, General Purpose and Special Equipment; Self-Propelled and Towed; General Specification for

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 601—Rubber: Sampling and Testing

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–417 —Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests)

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM METHOD D15–58T —Sample Preparation for Physical Testing of Rubber Products

ASTM METHOD D518–57T —Resistance to Surface Cracking of Stretched Rubber Compounds

ASTM METHOD D1149–60T —Test for Accelerated Ozone Cracking of Vulcanized Rubber

ASTM METHOD D1646–59T —Viscosity and Curing Characteristics of Rubber by

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the Shearing Disk Viscometer

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 **Qualification.** The unvulcanized rubber stock furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 4.3).

3.2 **Rubber materials.**

3.2.1 *Style I block.* The polymer shall be 100 percent cold processed SBR synthetic rubber or other synthetic polymers containing no reclaimed materials. The polymer shall cure in a maximum of 120 minutes, at a mold surface temperature of approximately 320° F. and under pressure sufficient to close the mold.

3.2.2 *Style II pin bushing.* The polymer shall be 100 percent natural rubber or blends of it with synthetic polymers containing no reclaimed materials. The polymer shall cure in a maximum of 45 minutes time, at a mold surface temperature of approximately 320° F. and under sufficient pressure to close the mold.

3.3 **Design, dimensions, and construction.** The extruded rubber stocks shall conform to the dimensional and weight requirements as specified on the applicable drawings (see 6.2) and as specified herein. If, in the event a conflict exists between the dimensional requirements and the weight requirements of the drawing, the dimensional requirements shall take precedence.

3.4 **Processability.** The unvulcanized rubber stocks shall be capable of being processed in rebuilding equipment, utilizing the methods of the facility performing the vulcanization and assembly for reconditioning track shoes and blocks.

3.5 **Storage life.** The unvulcanized rubber stocks shall withstand a minimum storage period of 12 months after date of manufacture without showing more than a light sulfur bloom or any evidence of precuring and shall be capable, after storage under the conditions listed below, of being processed the same as the new material:

(a) Protection from physical damage during storage period.

(b) The ambient storage air temperature has not exceeded 90° F.

3.5.1 *Accelerated storage life.* In lieu of the twelve months storage (see 3.5), the unvulcanized stock shall withstand oven aging at 158° F. for 7 days as determined by the Mooney Scorch test (See 4.6.2). During the oven aging cycle the test pieces shall be completely inclosed in cellophane and suspended in the oven.

3.6 **Ozone resistance.** The unvulcanized rubber stock shall be such that after vulcanization into track components, except bushings, it shall be ozone resistant and when tested as specified in 4.6.4 shall evidence no cracks under 7 power magnification.

3.7 **Adhesion.**

3.7.1 *Style I stock.* When molded into a style I block within the limits specified in 3.2.1 and tested as specified in 4.6.1, the load required to separate the rubber from the metal of the type I shoes and pads, and from the tubes of the type II shoe frames shall be no less than 150 pounds. The load required to separate the rubber from the end links of the type II frames shall be no less than 60 pounds (fig. 1–4).

3.7.2 *Style II stock.* When molded into a style II pin bushing within the limits specified in 3.2.2 and when tested as specified in 4.6.1.4, the load required to separate the rings from the pin shall be not less than that shown in table I. The rubber pulled off and that remaining on the pin shall show no evidence of blistering or porosity at the conclusion of the test.

3.8 **Rubber physical properties.** When tested as specified in 4.6, the rubber bushing specimens (see 4.6.2.1) and the rubber in track shoes and pads shall conform to requirements of table II.

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TABLE I. *Load to separate rubber from pin*

Diameter of metal pin (inches)	Load, (pounds per inch of width) minimum
1	29
1⅛	31
1¼	33
1⅜	34
2	36
2⅜	38

Note. When exact size of pin is not shown in table, use value of load for next larger size.

3.9 Bushing functional properties.

3.9.1 *Load deflection.* When rubber bushing specimens (see 4.6.2.1) are tested as specified in 4.6.3.1, the load deflection shall be from 33 to 41 percent.

3.9.2 *Compression set.* When rubber bushing specimens (see 4.6.2.1) are tested as specified in 4.6.3.2, the compression set shall not exceed 25 percent.

3.10 Low temperature resistance. When the rubber from the bushing specimens, shoes, and pads is tested as specified in 4.6.5 at a temperature as specified in table II, the specimen shall show no evidence of cranks, fissures, or pinholes.

TABLE II. *Physical properties*

Physical property	Bushings	Shoes and pads
Hardness: Shore A durometer	55–68	63–73
Tensile strength:		
Before aging p.s.i., min	3500	2900
After aging (70 hrs. at 158° F.) psi, min	3000	
After aging (70 hrs. at 212° F.) psi, min		2500
Elongation:		
Before aging percent, min	500	400
After aging (70 hrs. at 158° F.) percent min	425	
After aging (70 hrs. at 212° F.) percent min		300
Tensile stress:		
Before aging, at 400 percent elongation, psi, min	1800	
Before aging, at 300 percent elongation, psi, min		2000
Tear resistance, per inch of thickness:		
At 72° ± 2° F., pounds, min	(*)	300
At 250° ± 2° F., pounds, min	(*)	175
Specific gravity, maximum	1.16	1.21
Low temperature resistance	—67° F.	—40° F.
Mooney scorch resistance:		
Small rotor and at 250° F.		
Initial reading, max	75	75
Four minute reading, max	45	45
Minutes to 5 point rise, max	50	75
Minutes to 5 point rise, min	18	35
After aging 7 days at 158° F.:		
Initial reading, max	125	110
Four minute reading, max	80	65
Minutes to 5 point rise, max	60	75
Minutes to 5 point rise, min	15	30

* Not applicable

3.11 Road test. After the break-in run (see 4.6.6.1), when tested as specified in 4.6.6.2, track shoe assemblies and sets shall perform for the number of miles shown and under the conditions specified in table III, without evidence of excessive wear or such failure as would cause the track shoes and pads to become unserviceable. The average height of the rubber chevrons or pads at end of the road

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test shall be not less than 50 percent of their original height. Not more than 30 percent of the number of shoes or pads shall lose more than 3 square inches of rubber bearing area, and the original total bearing area of rubber chevrons or pads shall be reduced by not more than 7 percent.

TABLE III. *Miles and speed of vehicle*

Vehicle weight (pounds)	Average Speed (m.p.h.)	Distance (miles)
Under 60,000	25	2000
60,000 to 100,000	20	1500
Over 100,000	15	1000

Note. The speed may exceed the specified average speed only for short periods.

3.12 **Workmanship.** Manufacturing methods shall be in accordance with the applicable provisions of MIL–V–45349. Rubber in shoes and pads shall be free from blisters, cracks, folds, backrinding, flash, cuts, and non-fill (aircheck) (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Contractor's inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of tests.** The inspection and testing of the rubber stock shall be classified as follows:

(a) Qualification tests
(b) Acceptance tests
(c) Control tests

4.3 **Qualification tests.** For qualification tests the sample submitted shall consist of the following:

(a) 5 pounds of unvulcanized bushing stock in mill sheet form.
(b) 5 pounds of unvulcanized block stock in mill sheet form.
(c) Unvulcanized rubber stock extruded to conform to specified drawings (see 6.2) as applicable, in the following amounts:
    110 pieces—Track block, ground side.
    110 pieces—Track block, wheel side.
    440 pieces—Pin bushing.
    The samples shall be examined for defects specified in 4.4 and after vulcanization to their intended use, subjected to the applicable tests specified in table IV.
(d) Processing instructions. The instructions shall include the curing temperature and curing time within the limits as specified in 3.2.

4.3.1 *Failure to qualify.* Failure of any qualification test sample to pass the specified tests may be cause for refusal by the Government to conduct additional tests until it has been proved, to the satisfaction of the qualifying activity, that the faults revealed by the tests have been corrected.

4.4 **Acceptance inspection.**

4.4.1 *Sampling for unvulcanized rubber stock.*

4.4.1.1 *Lot formation.* Unless otherwise specified, a lot shall consist of all unvulcanized rubber stock of one style, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 *Sampling for acceptance examination.* Samples for acceptance examination shall be selected in accordance with MIL–STD–105, using single sampling.

4.4.1.3 *Sampling for acceptance testing.* Samples for acceptance testing, from lots that have passed the examination specified in 4.4.2, shall be selected in accordance with inspection level L5 of MIL–STD–105.

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TABLE IV. *Applicable tests*

Test	Requirement paragraph No.	Test paragraph No.	Qualification	Control	Acceptance
Ozone resistance					
Whole specimens	3.6	4.6.4.1	X		
Prepared specimens	3.6	4.6.4.2	X	X	
Adhesion: Type I shoe	3.7.1	4.6.1.1	X	X	X
Adhesion: Type I pad	3.7.1	4.6.1.2	X	X	X
Adhesion: Type II shoe	3.7.1	4.6.1.3	X	X	X
Adhesion: Bushing	3.7.2	4.6.1.4	X	X	X
Physical properties	3.8	4.6.2	X	X	
Bushing load deflection	3.9.1	4.6.3.1	X	X	
Bushing compression set	3.9.2	4.6.3.2	X	X	
Low temperature resistance	3.10	4.6.5	X	X	
Road test	3.11	4.6.6	X		

4.4.2 *Examination inspection.*

4.4.2.1 *Acceptable quality levels.* Each sample selected in accordance with 4.4.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	*AQL*
Major	1.0
Minor	2.5

4.4.2.2 *Classification of defects.* For the purpose of examination inspection, defects shall be classified as indicated below:

Categories	Defects	Inspection method
Major:		
101	Dimensional nonconformance (see 3.3).	Visual and gage
102	Weight not as specified (see 3.3).	Scale

4.4.3 *Testing inspection.*

4.4.3.1 *Acceptance tests.* Samples selected in accordance with 4.4.1.3 shall be subjected to the applicable tests specified in table IV using an AQL of 6.5 percent defective.

4.4.4 *Sampling for packaging and packing.*

4.4.4.1 *Lot formation.* A lot shall consist of all packs prepared for shipment in accordance with one level (see 6.2), from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.4.2 *Sampling for acceptance examination.* Sampling for acceptance examination shall be performed in the manner specified in 4.4.1.2.

4.4.4.3 *Sampling for acceptance testing.* One complete pack shall be selected for acceptance testing.

4.4.5 *Examination inspection for packaging and packing.* Acceptance quality levels and classification of defects shall be as specified by the procuring activity (see 6.2).

4.4.6 *Testing inspection for packaging and packing.*

4.4.6.1 *Acceptance testing.* Unless otherwise specified, the sample selected in accordance with 4.4.4.3 shall be tested as specified in 4.7.2 and 4.7.3.

4.4.6.2 *Failure* Failure of the sample to pass any specified test, may be cause for the Government to refuse to accept the lot until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 **Control tests.** Specimens for control testing shall be selected from the initial production run and from every other months production. Specimens shall be as follows:

(a) Two track blocks.

(b) Two pins with rubber bushings thereon.

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(c) Specimens from bushing stock, consisting of the following:
  (1) Three compression set buttons.
  (2) Three load deflection buttons.
  (3) Six 6 by 6 by .080 ± .005 inch test slabs.

The applicable specimens shall be subjected to the applicable tests specified in table IV, using an AQL of 6.5 percent defective. Unless otherwise specified in contract or order, specimens shall be submitted for testing to Commanding General, Headquarters, United States Army Tank Automotive Center, Detroit Arsenal, Centerline, Mich., ATTN: SMOTA–RCM.

4.5.1 *Control test failure.* Upon failure of a specimen to pass any test specified herein, the Government inspector may refuse acceptance of subsequent lots until objective evidence has been provided that corrective action has been taken.

4.6 Tests and test procedures.

4.6.1 *Adhesion (see 6.5).* One specimen shall be subjected to this test. The specimen shall be prepared for testing as shown in figure 1, for type I shoes; figure 2 or 3, as applicable, for type II shoes; and figure 4, for type I detachable pads. Results of this test shall meet the applicable requirements specified for adhesion are deemed impractical, such alternative methods as may be proposed by the contractor and approved by the Government contracting officer may be used to determine the adequacy of the rubber to metal bond.

4.6.1.1 *Type I shoes.* Strips, each 1 inch wide, shall be cut along the center of the shoe and parallel to the pins as shown in figure 1. With the shoe mounted in a holding fixture, arranged to slide longitudinally, one strip shall be started at the center of the shoe and pulled towards the outer end. The adhesion shall be measured on each side of the shoe.

4.6.1.2 *Type I detachable pads.* A strip 1 inch wide shall be cut diagonally across the pad as shown in figure 4. With the pad mounted in a holding fixture, the strip shall be started at one end and pulled to measure the adhesion to the metal base plate.

4.6.1.3 *Type II shoes.*

4.6.1.3.1 *Shoes with rubber chevron.* Strips, each 1 inch wide, shall be cut longitudinally above and parallel to the cylindrical parts of the metal frame as shown in figure 2 or 3, as applicable, and then cut transversely at the center. With the shoe mounted in a holding fixture, one strip shall be started near the end of the shoe and pulled toward the center, and the other strip started at the center and pulled toward the end so as to include measurement of adhesion over the end link of the shoe.

4.6.1.3.2 *Shoes with steel grousers.* A strip 1 inch wide shall be cut longitudinally above and parallel to one of the cylindrical parts of the frame as shown in figure 2 or 3, as applicable, and shall be cut transversely near one end of the shoe. With the shoe mounted in a holding fixture, the strip shall be started and pulled to the end so as to include measurement of adhesion over the end link of the shoe.

4.6.1.4 *Bushings.*

4.6.1.4.1 *Apparatus.* A tension testing machine or apparatus having a power-actuated clamp shall be used to apply and measure the force required to strip the rubber from the pin to which it is adhered. The clamp shall travel at a rate of 2 inches per minute. A hub and spindle assembly shall be provided for mounting the bushings and permitting full rotation about a fixed axis.

4.6.1.4.2 *Procedure.* This test shall be made on the outer rings and on two of the rings near the center portion of the bushing. The bushing shall be prepared for testing by cutting the rubber rings transversely down to the metal pin and separating circumferentially by hand or use of a knife, for a distance sufficient to permit a firm grip by the free end of the power-actuated clamp. The bushing shall then be mounted on the hub and spindle assembly with the free end of the rubber firmly gripped by the machine clamp. The machine shall be

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started and the rubber pulled circumferentially from the pin. Force measurements shall be recorded and observation made throughout the test to determine conformance to 3.7.2 and table I. The rubber pulled off the pin and that remaining shall be examined for conformance to 3.7.2.

4.6.2 *Physical properties.* To determine conformance to 3.8 and table II, physical property values shall be determined as specified in the applicable method of Standard Fed. Test Method Std. No. 601 (see table V), except for the Mooney scorch test which shall be in accordance with ASTM method D-1646.

4.6.2.1 *Rubber bushing specimens.* The specimen shall be prepared as specified in ASTM method 4-15, and shall be standard laboratory slabs of an equivalent state of cure to that of the rubber bushings. The slabs shall be prepared from uncured molding blanks of the bushing compound submitted for qualification or acceptance testing. The contractor shall submit evidence acceptable to the Government inspector that the slabs are of a state of cure equivalent to that of the bushings being submitted for qualification or acceptance.

TABLE V. *Physical properties test methods*

Physical property	Test method
Hardness	3021
Tensile strength	4111
Elongation	4121
Tensile stress	4131
Accelerated aging	7221
Specific gravity	14011

4.6.3 *Bushing functional tests.*

4.6.3.1 *Deflection.* The bushing specimen (see 4.6.2.1) shall be 0.250± .005 inch thick and 0.700 ± .005 inch in diameter. The thickness of the test specimen shall be measured with a dial micrometer. The specimen shall then be placed on the lower platen of a weight deflection machine. A load of 600 p.s.i. shall be applied for a period of 5 seconds, released for a period of 5 seconds, applied a second time for a period of 5 seconds and again released. After this preliminary conditioning treatment, the specimen shall be recentered on the lower platen and the final load of 600 p.s.i. applied at a uniform rate of 0.5 inch per minute. The base of the apparatus shall be tapped sharply several times to secure maximum deflection and to establish equilibrium. The compressed height of specimen shall then be recorded. Load deflection based on thickness of specimens prior to loading shall be calculated as follows to determine conformance to 3.9.1.

$$\frac{(\text{Original thickness} - \text{compressed height})}{\text{Original thickness}} \times 100 = \text{Percent deflection at 600 p.s.i.}$$

4.6.3.2 *Compression set.* To determine conformance to 3.9.2, the bushing specimen (see 4.6.2.1) shall conform to and be subjected to the compression set test, method 3311 of Standard Fed. Test Method Std. No. 601.

4.6.4 *Ozone resistance.* To determine conformance to 3.6, the rubber materials shall be tested as follows.

4.6.4.1 *Whole specimens.*

4.6.4.1.1 *Type I shoe or pad and type II shoe.* The whole track block shoe or pad of the type furnished for qualification test shall be tested as follows.

4.6.4.1.2 *Procedure.* The whole specimen, after a minimum of 7 days aging after vulcanization, shall be placed in ozone chamber and exposed for 7 days at a temperature of 100° F. ± 2° F. in an air ozone mixture containing 50 ± 5 parts by volume of ozone per hundred million parts of air.

4.6.4.2 *Prepared specimens.*

4.6.4.2.1 *Specimens.* The size of the specimens shall be in accordance with procedure B of ASTM method D518, except the thickness shall be .080 ± .005 inch. Specimens shall be obtained from either of the following:

(a) Three specimens shall be taken from track block chevron and shall not include any outer surface.

(b) Three specimens shall be taken from production molding blanks of uncured rubber compound, being used to rubberize track compo-

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nents, prepared in the form of ASTM tensile slabs and certified of an equivalent state of cure as the production molding.

4.6.4.2.2 *Procedure.* The specimen shall be tested in accordance with procedure B of ASTM method D518 and ASTM method D1149, except the length of the clamping strips shall be such as to facilitate placement within the ozone test chamber. Specimen shall then be placed in chamber and exposed for 7 days at a temperature of 100° ± 2° F. in an air ozone mixture containing 50 ± 5 parts by volume of ozone per hundred million parts of air. The specimen surfaces shall be finished smoothly and of uniform thickness in accordance with Part C of ASTM method D–15.

4.6.5 *Low temperature resistance tests.* Specimens shall be subjected to the suffix F¹ test of MIL–STD–417 using the solenoid or motor driven impact apparatus conforming to method 5321 or 5311 of Standard Fed. Test Method Std. No. 601, to determine conformance to 3.10.

4.6.6 *Road test.*

4.6.6.1 *Break-in run.* The test specimens shall be broken-in prior to the road test by having the vehicle, on which they are installed, continuously operated for 50 miles on a smooth secondary road free of large size gravel or crushed stone, as shown in table VI.

TABLE VI. *Speeds and distances for break-in run*

Division of break-in run	Speed m.p.h.	Distance miles (min.)
a	10	10
b	15	15
c	20	20
d	20–30	5

4.6.6.2 *Test.* After the break-in run, the vehicle shall be operated under the load conditions specified in 4.6.6.2.1 and over terrain as specified in 4.6.6.2.2 until it has attained the minimum mileage shown in table III. On completion of the test the rubber shall be checked for wear against the requirements of 3.11.

4.6.6.2.1 *Load conditions.* During the test, the tracks shall be loaded to the gross weight specified for the vehicle involved, with the load distributed approximately equal on each track when on a level surface, and the vehicle operated at a speed not in excess of the maximum specified for the vehicle.

4.6.6.2.2 *Terrain.* One-third of the total mileage shall be attained with the vehicle being operated on paved roads (concrete, asphalt, or macadam); one-third, on gravel composed mostly of small stones; and the other one-third, over cross-country terrain. During the test, the vehicle shall not traverse more than 150 continuous miles over any single type of terrain.

4.7 Inspection of preparation for delivery.

4.7.1 *Materials and processes.* The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery to determine conformance to requirements of Section 5, and specifications as referenced therein. Any deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

4.7.2 *Packaging test.* To determine conformance to 5.1.2 the package shall be subjected to the test specified for the method of MIL–P–116 as specified in the applicable packaging data sheet.

4.7.3 *Packing test.* To determine conformance to 5.1.3, the shipping container shall be subjected to the interior packing test of Specification JAN–P–100.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for all levels of protection specified shall be in accordance with the following.

5.1.1 *Cleaning.* Unvulcanized rubber stock shall be cleaned in accordance with Method C–1 of MIL–P–116.

5.1.2 *Packaging.* Unvulcanized rubber

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stock shall be packaged in accordance with Method IC–1 of MIL–P–116 as follows:

Immediately after cleaning, one of each unvulcanized rubber stock item shall be placed in a 4 mil, polyethylene bag, heat sealed, conforming to Type I of Specification L–P–378. The bag size shall be calculated as specified in MIL–P–14232.

5.1.2.1 *Unit package.* The packaged rubber stock shall be placed in a container conforming to MIL–B–43014. Each container shall contain the quantity of rubber stock required for one track shoe. Container shall be of sufficient size as to prevent distortion or deformation of the rubber stock items.

5.1.3 *Packing.* Unit packages shall be packed in shipping containers as specified in MIL–P–14232. Strapping of containers shall be in accordance with the applicable container specification and in accordance with MIL–P–14232.

5.1.4 *Marking.* Each unit package and shipping container shall be marked in accordance with MIL–P–14232 and include at least the following:

Title, number, and date of this specification.
Title, number, and date of applicable drawing.
Manufacturer's name and address.
Contract number.
Date of manufacture.
Pieces contained in container.
Gross weight of container.

5.1.4.1 *Special marking.* Each shipping container shall be marked with a label 3 by 3 inches, with adhesive back, bearing the following legend in white letters not less than ¼ inch high on a red background with a white border not less than ⅛ inch wide:

"STORE IN COOL DRY PLACE."

6. NOTES

6.1 Intended use. The unvulcanized rubber stock covered by this specification is intended for use in rebuilding track shoes of tracked military vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style of unvulcanized stock required.
(c) Title, number, and date of applicable drawings (see 3.3).
(d) Formation lot (see 4.4.1.1 and 4.4.4.1).
(e) AQL and defects in packaging and packing (see 4.4.5).
(f) Level of protection (see 5.1).

6.3 Definition of defects.

(a) Blister. Void or hole in item; causes protrusion on surface when hot; may not show on surface when cold; may be covered or open.
(b) Crack. Failure of stock to knit together properly.
(c) Nonfill. Air checks; depressions due to trapping of air at corners and edges.
(d) Light stock. Insufficient material to fill mold.
(e) Fold. Lapping or doubling of material; results in crease or pleat.
(f) Backrinding. Distortion at the mold line—usually in the form of wrinkles, folds, tears, or indentations.
(g) Flash rubber. Spew-out of material at mold parting line.
(h) Cut. Nick, slit, or gash.

6.4 Illustrations. The illustrations shown herein (fig. 1–4) are relevant to the preparation of rubber specimens for testing purposes; they are not intended to represent shoe or pad designs.

6.5 Adhesion. The quality of adhesion of the rubber to the metal parts is of the greatest importance in securing satisfactory service from the shoes and pads. The method employed to secure this adhesion is, however, subject to many variables in the manufacturing process. The frequency of inspection should therefore be increased, whenever such variables tend to affect the quality of adhesion, or the adhesion value is barely sufficient for acceptance of the shoes and pads (see 4.6.1).

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6.6 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and the manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from Commanding General, Headquarters, United States Army Tank-Automotive Center, Detroit Arsenal, Center Line, Michigan, ATTN: SMOTA–RSS.1.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—MO

Preparing activity:
Army—MO (ATAC)

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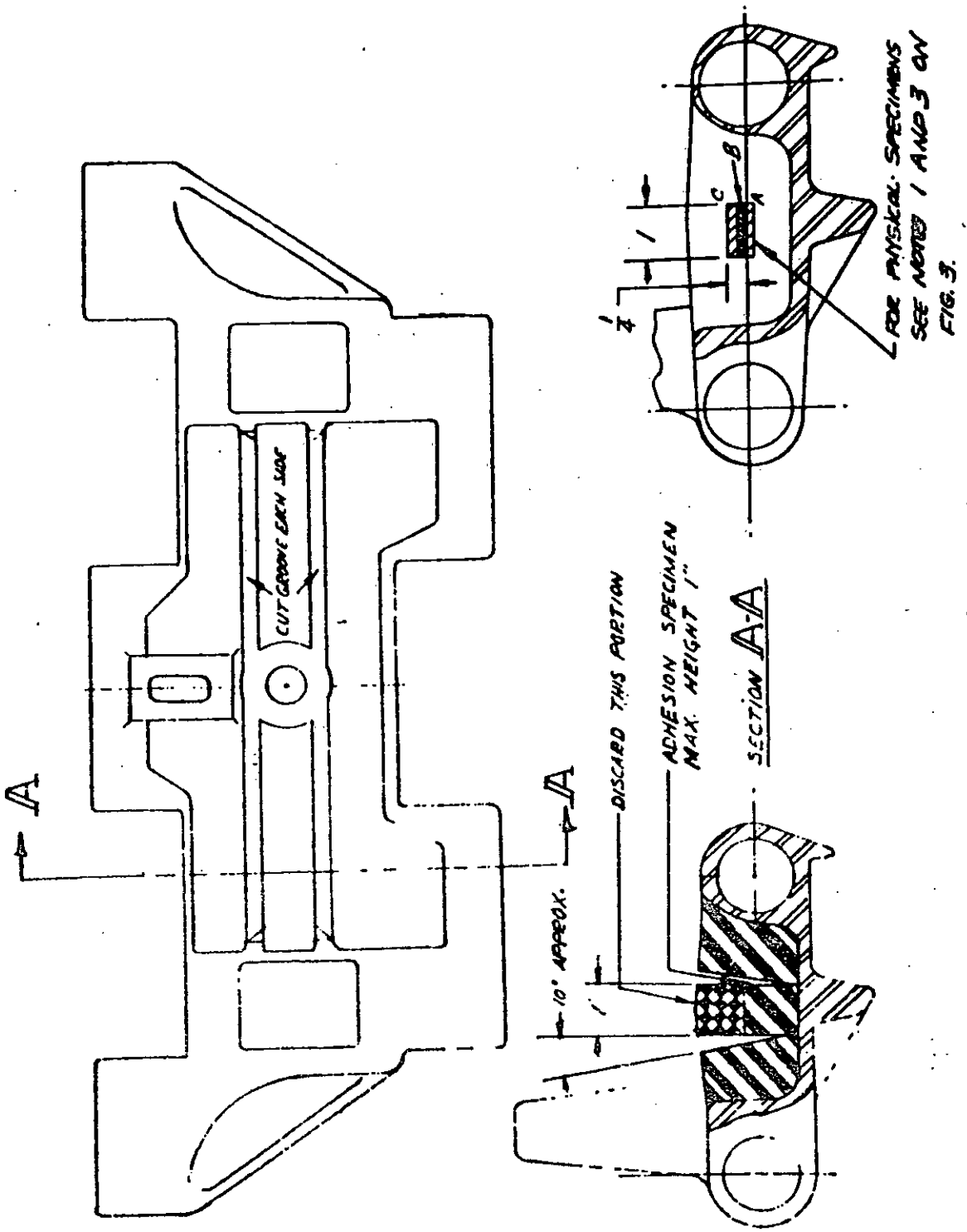
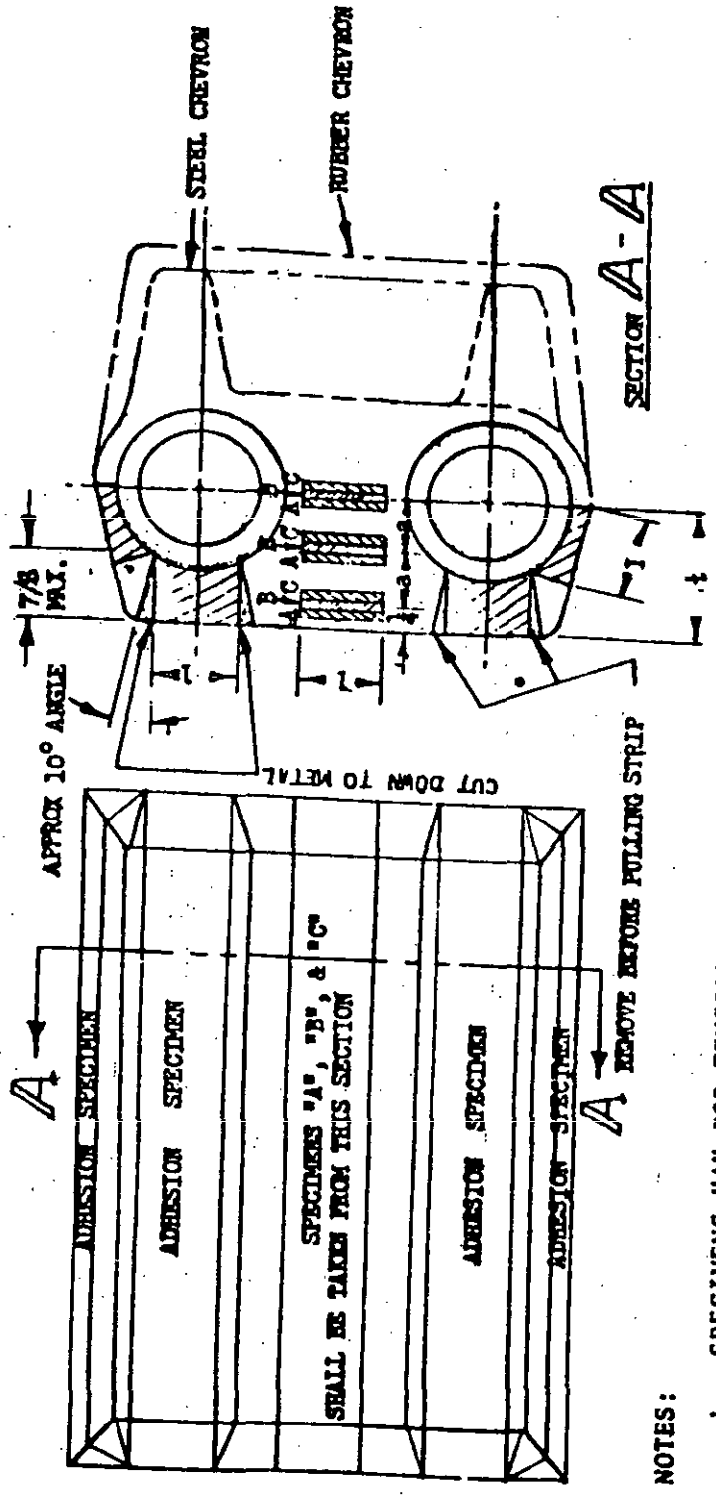


FIGURE 1. Type I. shoe test specimens.

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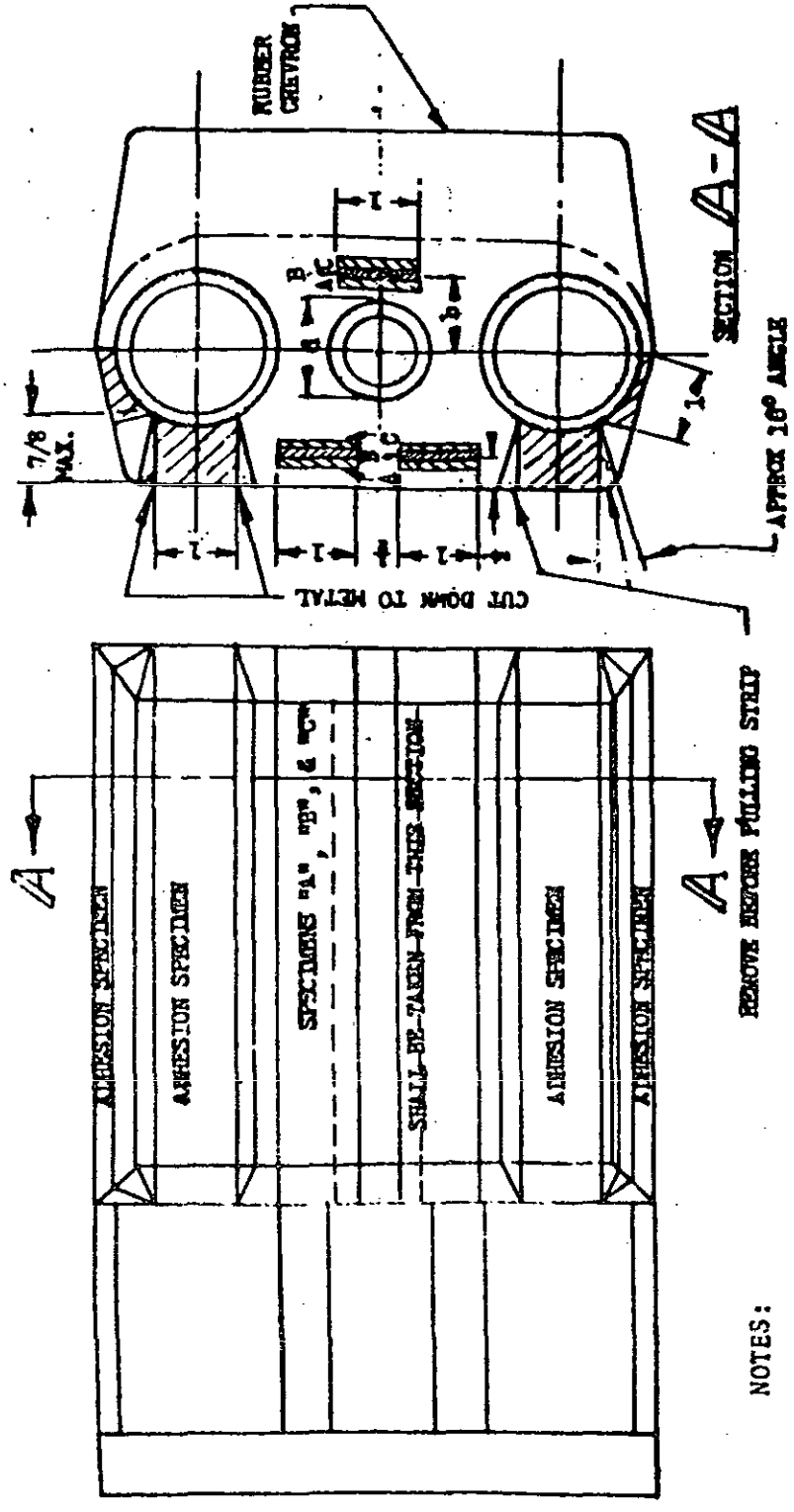


NOTES:

1. SPECIMENS "A" FOR TENSILE AND ELONGATION
   SPECIMENS "B" FOR AGING
   SPECIMENS "C" FOR ADDITIONAL SPECIMENS
2. a = (ℓ - 1/4)/2
3. STRIPS SHALL BE CUT TO END OF BLOCK ON BOGIE TIRE SIDE.

FIGURE 2. *Type II. shoe (two tube) test specimens.*

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NOTES:

1. SPECIMENS "A" FOR TENSILE AND ELONGATION
   SPECIMENS "B" FOR AGING
   SPECIMENS "C" FOR ADDITIONAL SPECIMENS
2. b = (d/2) + 1/4
3. STRIPS SHALL BE CUT FULL LENGTH.

FIGURE 3. *Type II. shoe (three tube) test specimens.*

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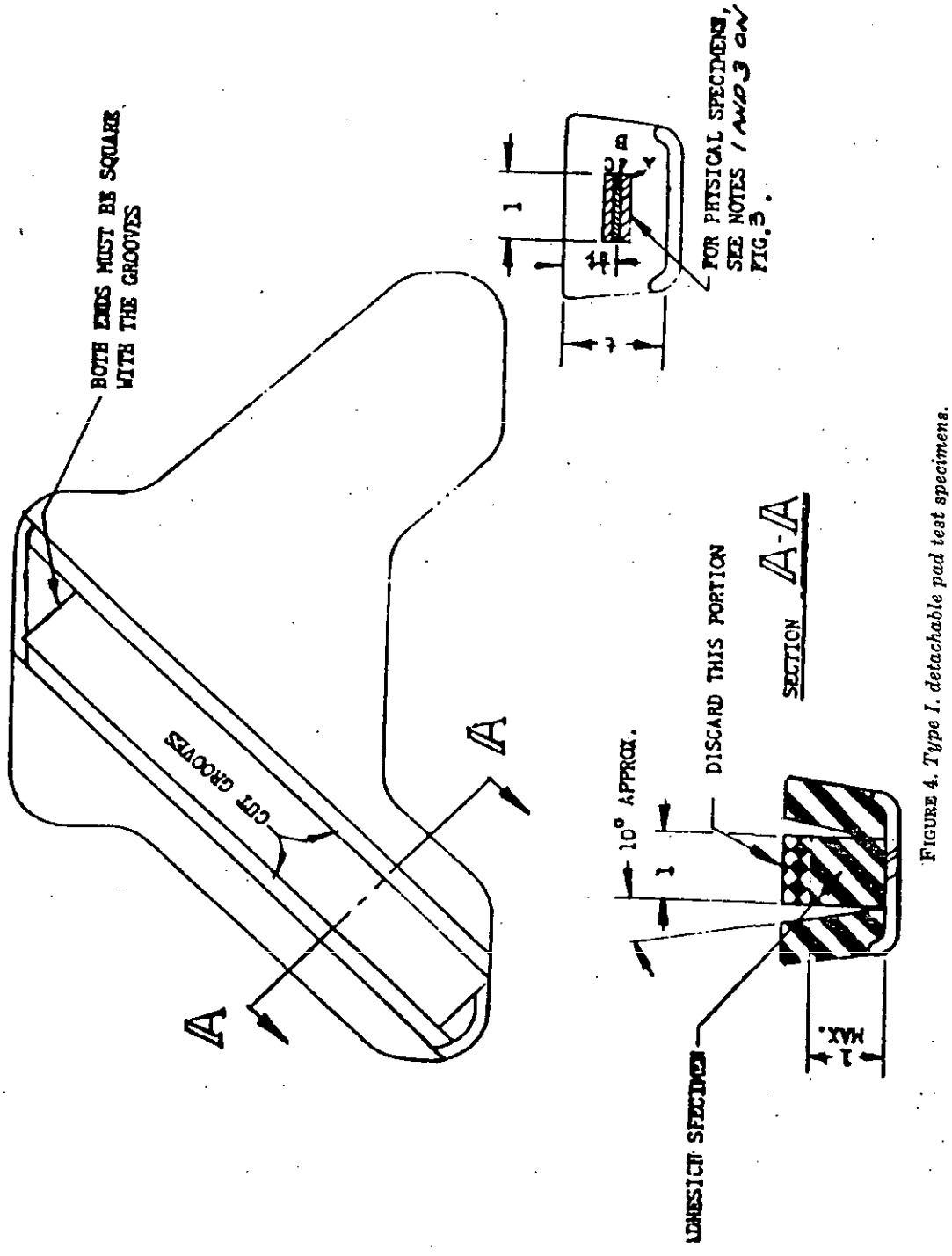


FIGURE 4. *Type I. detachable pad test specimens.*

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:PARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING GENERAL
U.S. ARMY TANK AUTOMOTIVE CENTER
ATTN: SMOTA-RSS
WARREN, MICHIGAN 48090

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM [illegible]

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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